Macroeconomics #0

Introduction

Quang-Thanh Tran

Postdoctoral Fellow, Tohoku University
Fellow, DEPOCEN

CASED Pre-Master Macroeconomics
October 7, 2026

About Me

- Education: PhD in Economics from Tohoku University, Japan (2025/03)
- Current position:
  - Postdoctoral Fellow, Tohoku University (from November 2026)
  - Research Fellow at DEPOCEN (permanent)
- Research Interests:
  macroeconomics, economic growth, life-cycle dynamics, demographic economics, and monetary economics.
- Email: `quangthanhtran.econ@gmail.com`
- My website: thanhqtran.github.io

Why should you take this course?

- Smoother transition to graduate schools
- Understand macroeconomics better
- Discussion on jobs and career? Yes, please.

Some short history

Mainstream

1. Neoclassical growth and traditional Keynesian economics (1950s–1960s)
2. Lucas critique and new classical revolution (1970s)
3. Real Business Cycles as the first generation of DSGE (1980s)
4. New Keynesian as the second generation of DSGE (1990s–2000s)
5. Frictions, crises, and limits of RANK (1990s–2010s)
6. Heterogeneous agents as the third generation of DSGE (2010s–today)
7. Life-cycle modelling and quantitative macroeconomics (1980s–today)

More policy-oriented

1. Empirical macroeconomics for forecasting
2. Semi-structural macroeconomic models (IMF QPM, FRB/US, East European Central Banks)
3. CGE models (computable general equilibrium)

Motivation for you

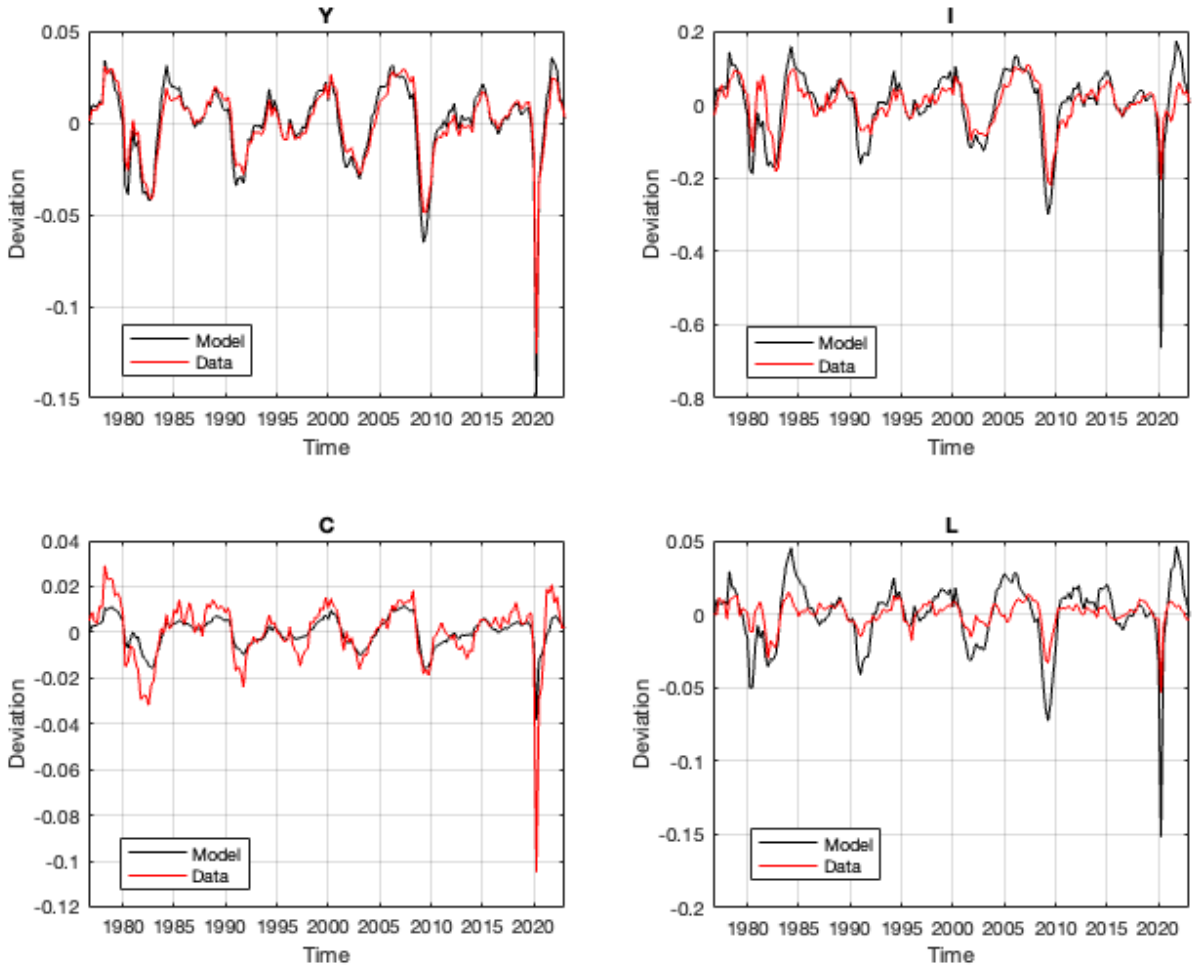


Figure: Application of Real Business Cycle on US Data to forecast GDP, Consumption, Investment, and Labor

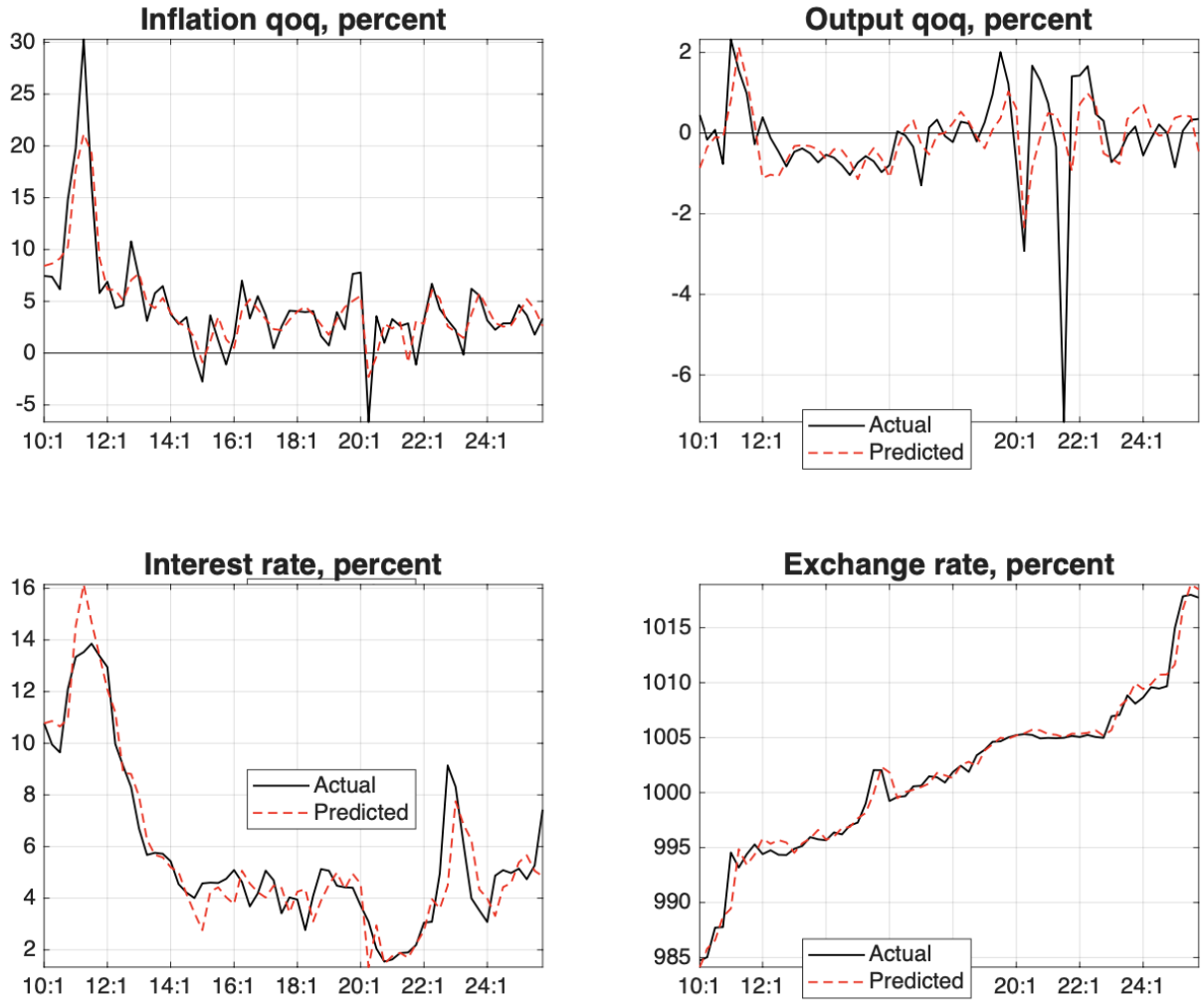


Figure: ... applied to Viet Nam to forecast inflation, GDP, interest rates, and exchange rate.

What will we study?

Content for each class

1. Some math tools
2. Solow Model
3. Ramsey Model
4. OLG Model
5. Endogenous Growth Model
6. Two-period Models and Richer OLG Dynamics
7. Real Business Cycle Model

My policies

1. Homework
2. Exam
3. The use of AI

Brief Discussions

A lot of people attack macroeconomics (or economics as a science as a whole), I am here to clear your doubts on our profession.

1. Math

   macroeconomics uses heavy mathematics as a "pseudo-science" to project false precision, cloak political ideologies, and mask the limitations of studying human behavior.

2. Models

   Most economic models rest on a number of assumptions that are not entirely realistic. Models are often a stylized version of reality, and there are so many realities.

3. Merits

   Due to the previous point, macroeconomics has no merit in informing policies.

Some books that will help you along the way

Math and dynamical analysis

- Sydsaeter, K., Seierstad, A. & Strom, A. (2008). Further mathematics for economic analysis. Pearson education.
- Simon, C. P., & Blume, L. (1994). Mathematics for economists. New York: Norton.
- Sorger, G. (2015). Dynamic Economic Analysis. Cambridge University Press.
- Shone, R. (2002). Economic Dynamics: Phase diagrams and their economic application. Cambridge University Press.

Dynamic Programming and Optimal Control

- Stokey, N. L., & Lucas Jr, R. E. (1989). Recursive methods in economic dynamics. Harvard University Press.
- Ljungqvist, L., & Sargent, T. J. (2018). Recursive macroeconomic theory. MIT press.
- Weber, T. A. (2011). Optimal control theory with applications in economics. MIT press.
- Kamien, M. I., & Schwartz, N. L. (2012). Dynamic optimization: the calculus of variations and optimal control in economics and management. courier corporation.

Suggested Books for This Course

- Chu, A. C. (2021). Advanced Macroeconomics: An introduction for undergraduates.
- Jones, C., & Vollrath, D. (2013). Introduction to Economic Growth. 3rd ed.
- Acemoglu, D. (2008). Introduction to modern economic growth. Princeton university press.
- Barro, R., & Sala-i-Martin, X. (2004). Economic growth second edition.
- Campante, F., Sturzenegger, F., & Velasco, A. (2021). Advanced macroeconomics: an easy guide. LSE Press.
- Romer, D. (2019). Advanced Macroeconomics. 5th Ed.
- de la Croix, D., & Michel, P. (2002). A theory of economic growth: dynamics and policy in overlapping generations. Cambridge University Press.
- McCandless, G. (2008). The ABCs of RBCs: An introduction to dynamic macroeconomic models. Harvard University Press.